Supporting Information

Probabilistic dengue and chikungunya forecasting in Brazil: metric-dependent model rankings, regime-shift fragility, and conformal ensembles

This document provides the scoring definitions (S1), full model configurations (S2), the conformal recalibration and ensemble-composition robustness analysis (S3), normalized WIS by season (S4), the EPIFORGE 2020 reporting checklist (S5), exploratory data analysis (S6), the reproducibility statement (S7), per-state model performance (S8), and a glossary of the technical terms used in the main text (S9). All numbers correspond to the committed backtest and are reproducible from the released pipeline.

S1 Text. Scoring

Forecasts are submitted as the nine quantiles $\tau \in \{0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.975\}$ per state and week, which define the median and the four central prediction intervals at nominal levels 50, 80, 90, and 95 percent. The weighted interval score is

$$\mathrm{WIS}(F,y) = \frac{1}{K+\frac{1}{2}}\Big[\tfrac{1}{2}\,|y-m| + \sum_{k=1}^{K} \frac{\alpha_k}{2}\,\mathrm{IS}_{\alpha_k}(\ell_k,u_k,y)\Big], \qquad \mathrm{IS}_\alpha(\ell,u,y) = (u-\ell) + \tfrac{2}{\alpha}(\ell-y)\mathbf{1}\{y<\ell\} + \tfrac{2}{\alpha}(y-u)\mathbf{1}\{y >$$

with m the predictive median, $K = 4$ central intervals, and $\alpha_k = 1 - L_k/100$. This equals the mean pinball loss over the nine quantiles up to the constant weighting, the exact discrete approximation to the continuous ranked probability score. We verified our implementation against the challenge platform's own scorer to machine precision on a shared set of forecasts, and against a hand-computed toy case that caught an early normalization error (division by 9 rather than $K + \frac{1}{2} = 4.5$): for median $m = 10$, a 50 percent interval $[6, 14]$, a 90 percent interval $[2, 20]$, and $y = 15$, the median term is $0.5|15 - 10| = 2.5$; $\mathrm{IS}_{0.5} = (14 - 6) + \frac{2}{0.5}(15 - 14) = 12$, weighted by $\alpha/2 = 0.25$ to 3; $\mathrm{IS}_{0.1} = (20 - 2) = 18$ (not exceeded, so no penalty term), weighted by 0.05 to 0.9; giving $\mathrm{WIS} = (2.5 + 3 + 0.9)/(2 + 0.5) = 2.56$ for this $K = 2$ toy case (the full score uses all four official intervals, $K = 4$). This score decomposes exactly into three non-negative components: a dispersion term (the interval-width contribution, present even when the median is exactly correct), an under-prediction penalty (positive only when y exceeds the interval's upper bound), and an over-prediction penalty (positive only when y falls below the lower bound); their sum is bit-for-bit equal to WIS by construction, which we also test. Alongside mean WIS we report: empirical interval coverage; that dispersion/under-prediction/ over-prediction decomposition of WIS; a per-unit relative WIS that scores each state-season-horizon against the naive baseline and takes a geometric mean, which weights every unit equally; and the normalized WIS, the sum of

WIS divided by the sum of observed cases, which is the challenge's headline metric and is scale-free across regions.

S2 Text. Model configurations

Features (gradient boosting). Both gradient-boosting implementations evaluated in the main text, LightGBM and XGBoost, are fit on the identical synthetic-origin panel that expands each fold into many weekly origins crossed with forecast horizons 1 to 67 weeks, with origin-anchored features computed only from the cutoff-filtered history:

- *Autoregressive:* log-incidence lags at 1, 2, 3, 4, 8, and 52 weeks; rolling mean over 4 and 8 weeks; rolling standard deviation over 4 weeks; rolling maximum over 12 weeks.
- *Seasonality:* target epidemiological week; first and second sine and cosine harmonics; a target-relative seasonal anchor equal to the most recent same-target-epiweek incidence at or before the origin.
- *Observed climate (four-week means at the origin):* temperature, precipitation, and relative humidity, plus a temperature anomaly against each state's same-epiweek climatology.
- *Seasonal climate forecast:* the seasonal forecast of temperature, humidity, and precipitation for the target month, merged leakage-safely so that only forecasts issued at or before the origin are used.
- *Ocean oscillation indices:* El Niño–Southern Oscillation, Indian Ocean Dipole, and Pacific Decadal Oscillation, each at lags 0, 4, 8, 12, 26, and 52 weeks.
- *Static:* log population; population-weighted Köppen climate-zone fractions; biome fractions; forecast horizon in weeks.

Hyperparameters (S1 Table). Table 1 lists the frozen configuration of each model. These were chosen a priori as conservative defaults. A separate search that preferred a deeper boosted configuration (num_leaves 63, max_depth 7) improved an internal time-block holdout by about 5 percent but worsened the true backtest, and was not adopted (main text the main text, negative results).

S3 Text. Conformal recalibration and ensemble composition

Conformal recalibration. For each central interval level L with bounds $[\ell_L, u_L]$ and median m, and for each calibration unit, the multiplicative factor needed to just cover the observation is $r = (y - m)/(u_L - m)$ when $y > m$ and $r = (m - y)/(m - \ell_L)$ when $y < m$, and 0 otherwise. The widening factor s_L is the $L/100$ empirical quantile of r over the tuning season, so that widening the interval about the median by s_L yields approximately L percent empirical coverage. The estimated factors are $s_{50} = 1.08$, $s_{80} = 1.04$, $s_{90} = 1.19$, and $s_{95} = 1.86$. They are gentle and concentrate on the outer tails, which is why the recalibration protects against catastrophic under-prediction in the outbreak season at small cost in ordinary seasons. Recalibrating the ensemble output outperforms recalibrating any single member, because the per-quantile median across members otherwise dilutes

Table 1: S1 Table. Frozen model configurations.

Model	Configuration
LightGBM (per quantile)	objective: quantile at each of the nine levels; num_leaves 31; max_depth 5; learning_rate 0.05; 400 boosting rounds; feature_fraction 0.8; bagging_fraction 0.8; bagging_freq 1; seed 42; deterministic. Intervals calibrated by conformalized quantile regression on a trailing 78-week window. Used for the chikungunya-state track; superseded by XGBoost below for the dengue ensemble.
XGBoost (multi-quantile, chosen for dengue)	objective: `reg:quantileerror` with `quantile_alpha` set to all nine levels jointly, `multi_strategy= multi_output_tree` (one booster covers all quantiles); `tree_method=hist`; max_depth 5; min_child_weight 30; learning_rate 0.05; 400 boosting rounds; subsample 0.8; colsample_bytree 0.8; seed 42. Same feature set and CQR calibration scheme as LightGBM above.
GRU deep ensemble	input sequence length 104 weeks; single-layer GRU encoder, hidden size 48, shared across all 26 states; per-state embedding of dimension 8; decoder is a two-layer perceptron with hidden size 64, ReLU, and dropout 0.1; negative-binomial output head; quantiles analytic from the predictive distribution; deep ensemble of 5 independently seeded runs pooled by quantile averaging; 40 epochs, Adam, learning rate 10^{-3}.
Semi-mechanistic	800 bootstrap resamples of whole historical season trajectories; season length 53 weeks; negative-binomial observation model; a local reimplementation of Richards-curve epidemic-scanner fits.
Ensembles	per-quantile median (Vincentization) of the climatological, gradient-boosted, and deep-learning models; and inverse-WIS weighting tuned only on the designated tuning season.

a member's widening (recalibrating the deep-learning member alone yields mean WIS 1273 versus 1162 for the ensemble-output recalibration).

Member-composition robustness (S2 Table). Table 2 scores every member subset under the same conformal recalibration. The chosen three-member ensemble is close to the best composition by the headline normalized WIS, and no subset is better on both the all-season and the ordinary-season metric. The subset that appears best by all-season mean WIS and normalized WIS (climatological plus mechanistic, 1141 and 0.551) is the worst of the strong subsets on ordinary seasons (normalized WIS 0.455 versus 0.375), because it wins only by handling the 2024 outlier and drops the two machine-learning models that carry the ordinary-season signal – the composition-level analogue of the metric-disagreement finding. It is also, by construction, an unreliable way to choose a composition: evaluating candidate subsets on the very seasons being reported invites exactly the overfitting the tuning-fold discipline exists to prevent. Scoring each candidate on fold 1 alone (the only fold this project's own discipline permits for such a decision) instead of on the full four-season table exposes this directly: the climatological-plus-mechanistic subset that looks best on the reported seasons is in fact one of the worst by that fold-1 criterion (normalized WIS 0.390, against 0.306 for the chosen composition and 0.282 for the single best-scoring subset, XGBoost plus GRU, which is itself the worst composition of all on the reported ordinary-season metric, 0.365 notwithstanding) – a within-study demonstration of why ensemble composition should be fixed by the tuning fold and not by inspecting the headline table.

Weight robustness. Inverse-WIS ensemble weights are stable across the choice of tuning season: tuning on fold 1 gives (climatological 0.32, XGBoost 0.33, GRU 0.35); on fold 2, (0.34, 0.37, 0.29);

Table 2: S2 Table. Ensemble composition sweep (each subset with conformal recalibration). nWIS is normalized WIS; nWIS ordinary excludes the 2024 outlier (fold 2). Fold-1 nWIS is the same quantity computed on the tuning fold alone, the only basis this project's discipline permits for choosing a composition; several subsets that look attractive on the reported columns fail this check. The chosen composition is in bold.

Members	**Mean WIS**	**nWIS all**	**nWIS ordinary**	**Fold-1 nWIS**
Climatological + Mechanistic	1141	0.551	0.455	0.390
Climatological + XGBoost + GRU (chosen)	**1162**	**0.561**	**0.375**	**0.306**
Climatological + GRU + Mechanistic	1170	0.565	0.421	0.372
Climatological + XGBoost + GRU + Mechanistic	1182	0.571	0.386	0.335
GRU + Mechanistic	1194	0.576	0.377	0.343
Climatological + GRU	1202	0.580	0.373	0.309
XGBoost + Mechanistic	1211	0.585	0.408	0.335
XGBoost + GRU	1245	0.601	0.365	0.282

Table 3: S4 Table. Dengue normalized WIS by season. Lower is better. Best in each column in bold.

Model	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**
Ensemble (conformal)	0.306	0.665	0.308	0.827
Ensemble (Vincentization)	0.307	0.721	0.304	0.809
Ensemble (inverse-WIS)	**0.284**	0.731	0.296	0.812
XGBoost	0.341	**0.653**	0.370	0.948
Mechanistic	0.425	0.680	0.421	0.636
Climatological	0.350	0.724	0.394	0.626
GRU	0.316	0.833	**0.247**	1.100
Seasonal-naive	0.454	0.776	0.444	0.595
Naive	0.659	0.773	0.806	**0.540**

on fold 3, (0.27, 0.29, 0.44). The unweighted median ensemble avoids this choice entirely and performs comparably, which is why it is preferred.

S4 Table. Normalized WIS by season

Table 3 gives normalized WIS by model and season. It shows the regime split numerically: the deep network is best on the ordinary seasons (fold 3, 0.247) but worst on the 2024 outlier (fold 2, 0.833), while XGBoost has the best outbreak-season normalized WIS of any single model or ensemble (0.653), narrowly ahead of the conformal ensemble (0.665). Fold 4 (2025–26) is only partially resolved, and its normalized WIS values are correspondingly larger and noisier for every model than the other three folds; they should be read as provisional.

S5 Text. EPIFORGE 2020 reporting checklist

This manuscript follows the EPIFORGE 2020 reporting guidelines for epidemic forecasting and prediction research (Pollett et al. 2021). Table 4 maps all 19 recommended items, in the guideline's own numbering and wording, to their location in the main text and this document. Locations are given by section name rather than section number, so that they remain correct if the main text is reordered. Two items warrant a note. Item 16 concerns version-stamping results published as a data object: we release no standalone data object, but the provenance manifest described in S7

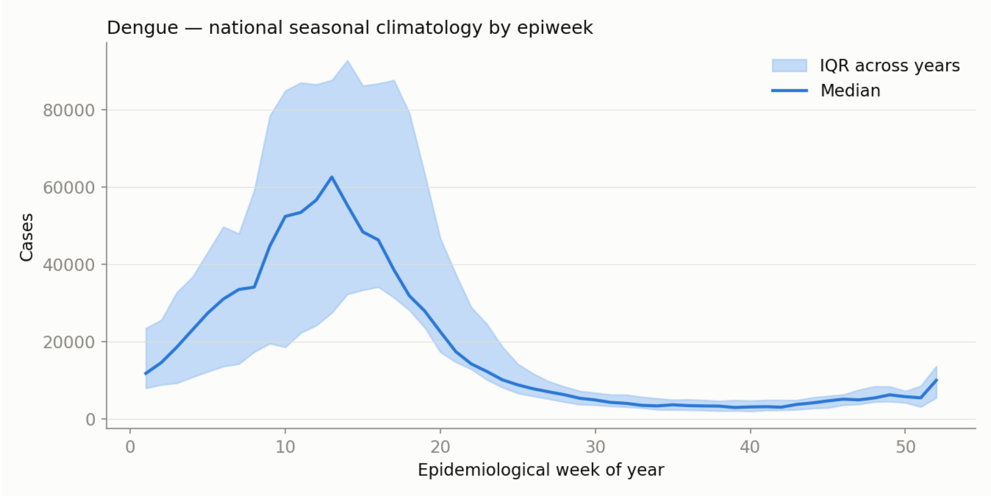

Figure 1: S1 Fig. Seasonal climatology of dengue incidence by state.

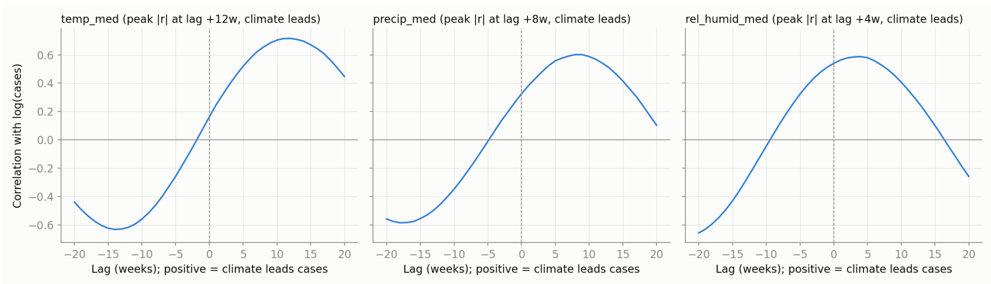

Figure 2: S2 Fig. Lagged cross-correlation of climate variables with dengue incidence.

Text serves the same purpose, recording the exact data checksums and code revision behind every reported number. Items 18 and 19 are conditioned on applicability to a specific epidemic; both apply here, and are addressed for the 2024 Brazilian dengue season specifically.

S6 Figs. Exploratory data analysis

The following figures summarize the data that motivated the modeling choices. S1 Fig shows the per-state seasonal climatology of dengue incidence. S2 Fig shows the lagged cross-correlation between climate variables and incidence. S3 Fig shows incidence stratified by Köppen climate zone. S4 Fig relates annual dengue totals to the El Niño–Southern Oscillation. S5 Fig shows the distribution of epidemic peak weeks across states and seasons.

S7 Text. Reproducibility and provenance

The complete pipeline is released as an installable package with a single command to reproduce every result. Leakage safety, scoring correctness, model persistence, and every model family are covered by an automated test suite that runs in continuous integration. All model fitting is deterministic under a fixed seed: LightGBM uses its own deterministic mode, XGBoost and the mechanistic model fix their own seeds, and the deep ensemble fixes the seed of each member. A provenance manifest records the exact data checksums and code revision behind each reported number, and the per-state scored predictions for all models are released so that any table or figure

Table 4: S3 Table. EPIFORGE 2020 checklist. All 19 recommended reporting items (Pollett et al. 2021), with the location of each in this manuscript.

#	Recommended item	Location
Overall study description and goals		
1	Describe the study as forecast or prediction research in at least the title or abstract.	Title; Abstract, first sentences; Author summary
2	Define the purpose of study and forecasting targets.	Introduction (contributions); Results, Study design and data
3	Fully document the methods.	Methods, all subsections; S1 and S2 Text
4	Identify whether the forecast was performed prospectively, in real time, and/or retrospectively.	Results, Study design and data (four retrospective seasons); Discussion, Limitations (the submitted 2026–27 forecast is prospective and not yet scored)
Data description		
5	Explicitly describe the origin of input source data, with references.	Methods, Data and targets; S2 Text
6	Provide source data with publication, or document reasons as to why this was not possible.	Data and code availability; S7 Text
7	Describe input data processing procedures in detail.	Methods, Data and targets; S2 Text (feature list)
Model characteristics		
8	State and describe the model type, and document model assumptions, including references.	Methods, Models; S2 Text; S1 Table
9	Make the model code available, or document the reasons why this is not possible.	Data and code availability; S7 Text
Model evaluation		
10	Describe the model validation, and justify the approach.	Methods, Backtest design and leakage control
11	Describe the forecast accuracy evaluation method used, with justification.	Methods, Scoring; S1 Text (formula, worked example, verification against the platform scorer)
12	Where possible, compare results to a benchmark or other comparator model, with justification.	Results, No single model dominates (naive and seasonal-naive baselines); Tables 1 and 2
13	Describe the forecast horizon, with justification of its length.	Methods, Backtest design and leakage control (16 to 67 weeks, set by the 15-week reporting gap and the EW41–EW40 target season)
14	Present and explain uncertainty of forecasting results.	Results, Calibration and its correction; Results, An ensemble is the most robust model; Methods, Conformal recalibration; S1 and S3 Text
Translation, interpretability, and generalizability		
15	Briefly summarize the results in nontechnical terms, including a nontechnical interpretation of forecast uncertainty.	Author summary
16	If results are published as a data object, encourage a time-stamped version number.	S7 Text (provenance manifest; see the note above)
17	Describe the weaknesses of the forecast, including weaknesses specific to data quality and methods.	Discussion, Limitations
18	If applicable to a specific epidemic, comment on potential implications for public health action and decision-making.	Discussion, Robustness under regime shift; Discussion, Conformal recalibration as transferable insurance; Results, Onset, peak, and total burden
19	If applicable to a specific epidemic, comment on how generalizable it may be across populations.	Discussion, Limitations (two diseases, one country, 26 states spanning the full incidence range)

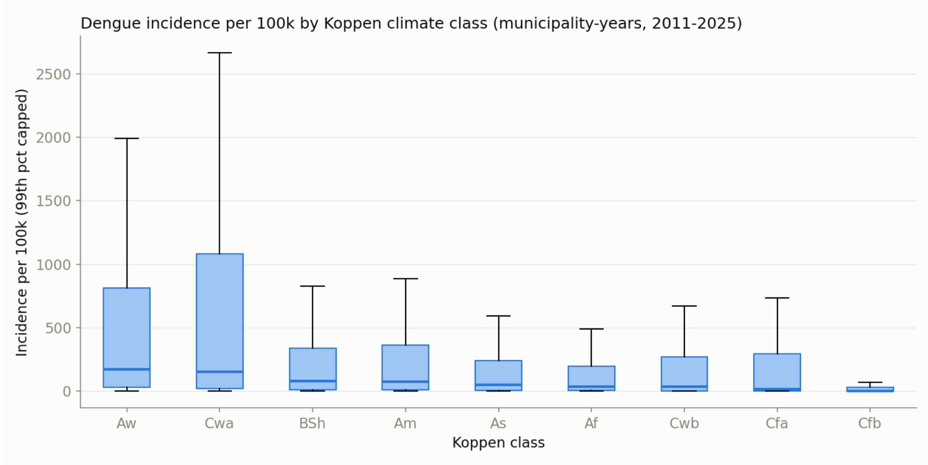

Figure 3: S3 Fig. Dengue incidence by Köppen climate zone.

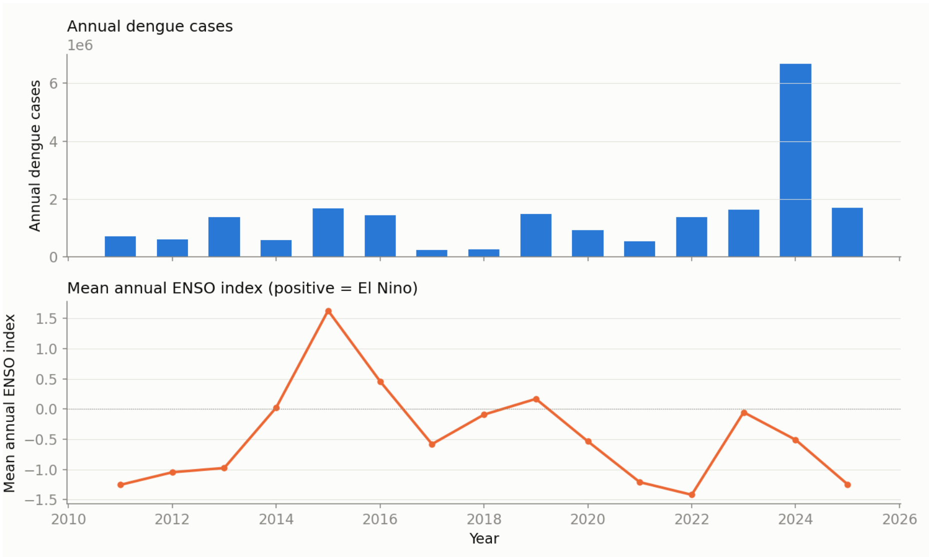

Figure 4: S4 Fig. Annual dengue totals versus the El Niño–Southern Oscillation index.

can be regenerated. Validated submission files for the four official tracks (dengue and chikungunya, at state and city level) are included.

S8 Text. Per-state model performance heatmap

Table-based per-state comparisons do not scale to six models across 26 states. Fig S6 gives the same comparison as a heatmap: each cell is one model's mean log relative WIS against the seasonal-naive baseline in one state, averaged over the four seasons, the same quantity underlying the stability figure in the main text (Fig 3) but broken out by state instead of collapsed to a mean and standard deviation. Green cells are better than the baseline and red cells are worse. Two patterns are visible beyond the aggregate numbers already reported: the deep network's advantage is concentrated in a subset of states (Ceara and Roraima most strongly) rather than uniform, and the gradient-boosted model (XGBoost) is the only one of the six with states that are red, meaning worse than the seasonal-naive baseline (Sergipe most strongly, followed by Paraíba, Amapá, Maranhão, Pará, Piauí, and Rondônia), which the state-level aggregate in Table 1 does not show.

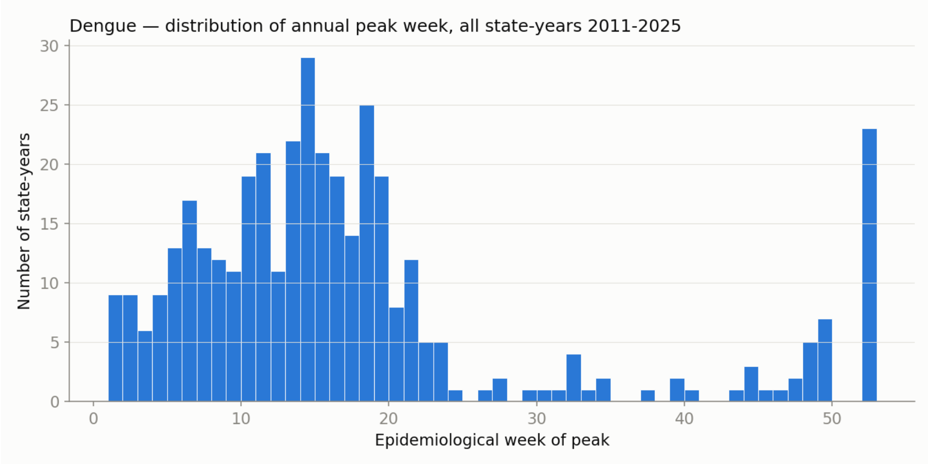

Figure 5: S5 Fig. Distribution of epidemic peak weeks across states and seasons.

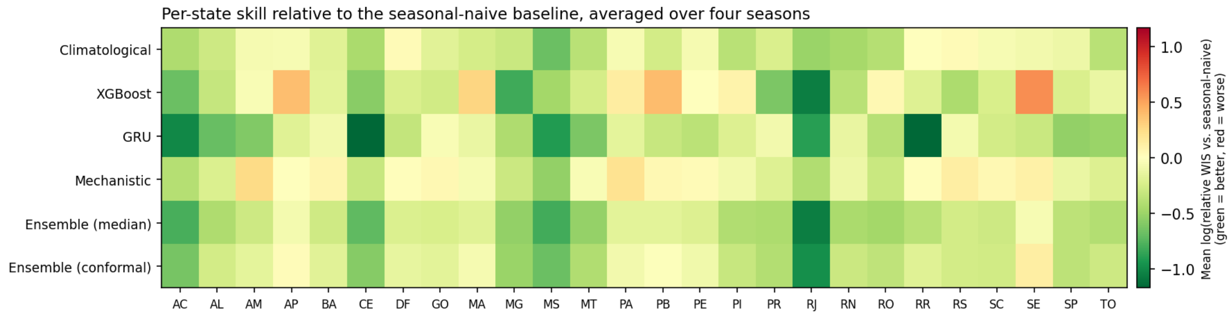

Figure 6: S6 Fig. Per-state skill relative to the seasonal-naive baseline. Mean log relative WIS by model and state, averaged over the four seasons. Green is better than baseline, red is worse.

S9 Text. Glossary

Terms used throughout the main text and this document, in order of first appearance.

- **WIS (weighted interval score)** — the challenge's scoring rule for a probabilistic forecast; combines median accuracy with interval width and coverage into one number, lower is better (S1 Text gives the formula and a worked example).
- **Coverage (nominal vs. empirical)** — nominal coverage is an interval's stated probability (e.g. 90 percent for the 90 percent interval); empirical coverage is the fraction of observations it actually contained. A well-calibrated model has the two close together.
- **Sharpness** — how narrow, and therefore informative, a model's prediction intervals are, independent of whether they are well calibrated.
- **Dispersion, under-prediction, over-prediction (WIS components)** — the three non-negative pieces WIS decomposes into: dispersion is the interval-width penalty present even for a perfectly centered forecast; the other two are penalties for the observation falling above or below the stated interval.
- **Relative WIS** — one model's WIS divided by a baseline model's WIS on the same forecasting units, a pairwise comparison (Cramer et al. 2022) that removes shared, unit-level variance; we report its logarithm so a model twice as good and twice as bad are symmetric.

- **Normalized WIS** — the sum of WIS divided by the sum of observed cases; scale-free across states/cities of very different size, and the challenge's own headline metric.
- **Vincentization** — combining several models' forecasts by taking, at every quantile level separately, the median of the models' values at that level.
- **Conformalized quantile regression (CQR)** — calibrating one model's own quantile predictions, during fitting, by measuring on held-out data how much each quantile needs to shift to reach its nominal coverage.
- **Conformal recalibration (this paper's ensemble-level step)** — a separate, after-the-fact widening of the already-combined ensemble's intervals, by a multiplicative factor per interval level tuned on one season and reused unchanged thereafter; distinct from CQR, which calibrates one model during fitting rather than the combined ensemble afterward.
- **Synthetic-origin panel** — the training set built for XGBoost/LightGBM/GRU: every historical week within a fold's cutoff-filtered history becomes a candidate forecast origin, crossed with horizons 1–67 weeks ahead, so a handful of true evaluation windows becomes tens of thousands of training rows.
- **Block bootstrap** — a resampling technique that repeatedly draws whole state-season units (not individual weeks) with replacement to estimate how much a summary statistic (e.g. normalized WIS) would vary under the data's actual dependence structure.
- **Paired comparison** — comparing two models' scores on the exact same forecasting units and looking at the difference, rather than comparing their two separate score distributions; cancels variance that is common to both models (e.g. from an unusually hard season).
- **Negative-binomial (head/distribution)** — a count-data distribution more flexible than Poisson, allowing variance to exceed the mean, as case-count data typically do; the GRU's output layer and the mechanistic model's observation model both use it.
- **Richards curve** — a flexible, S-shaped growth-then-plateau curve classically used to summarize an epidemic's rise and fall over a season; the mechanistic model fits one per historical season and resamples from these fits to build future trajectories.
- **Distributed-lag non-linear model (DLNM)** — a modeling structure that lets a covariate's effect on the outcome be both delayed (this week's climate affecting cases several weeks later) and non-linear in the covariate's own value.
- **Empirical-Bayes pooling (shrinkage)** — partially pulling each unit's (e.g. each state's) own estimate toward a shared average across a group of units, to borrow statistical strength when individual units are noisy.
- **Onset week, peak week/magnitude, total cases (operational summaries)** — decision-relevant quantities WIS does not directly show: the first week a series reaches a fraction of its own peak; the timing and size of the true peak week; and the season's total case count; each reported as an error relative to the true value (Results Section 2.11 of the main text).